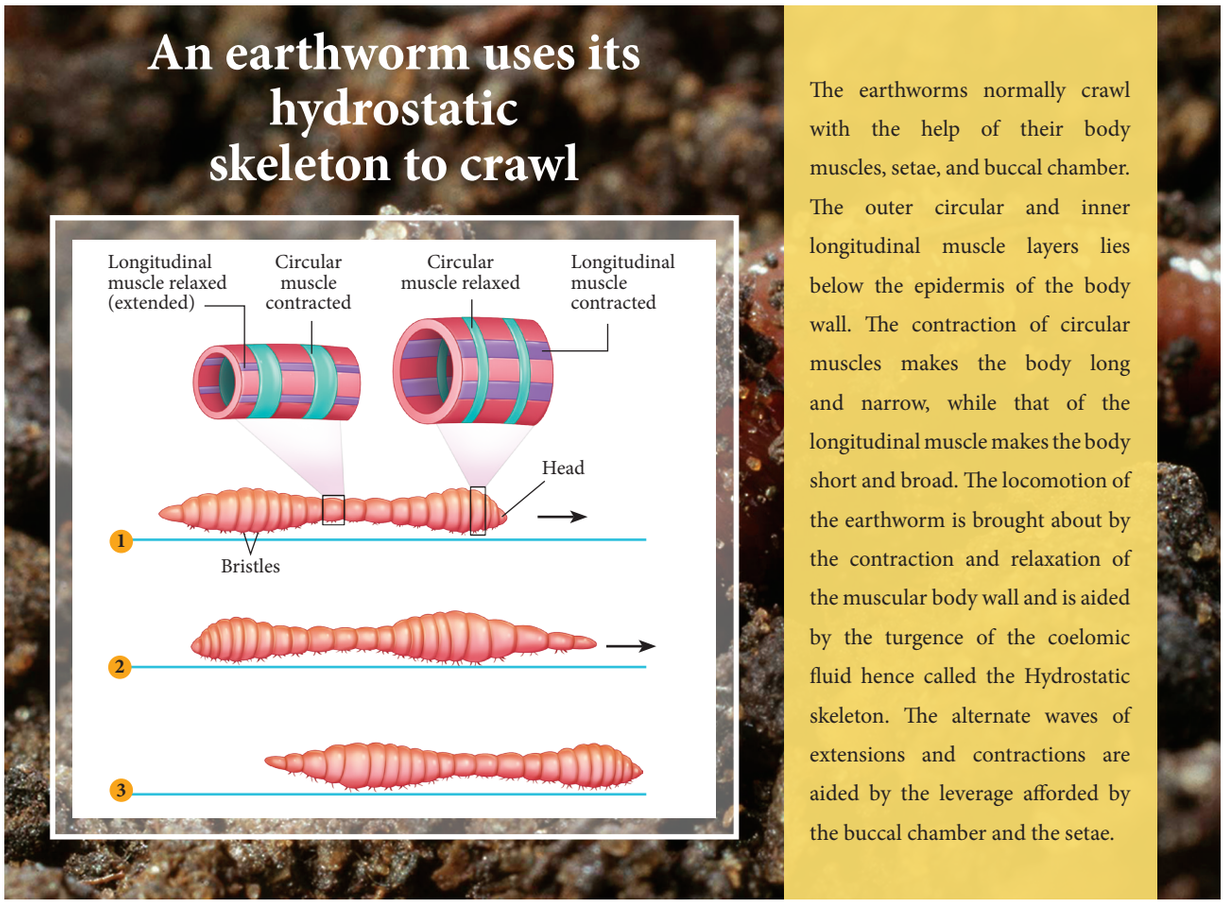


The earthworms normally crawl with the help of their body muscles, setae, and buccal chamber. The outer circular and inner longitudinal muscle layers lies below the epidermis of the body wall. The contraction of circular muscles makes the body long and narrow, while that of the longitudinal muscle makes the body short and broad. The locomotion of the earthworm is brought about by the contraction and relaxation of the muscular body wall and is aided by the turgence of the coelomic fluid hence called the Hydrostatic skeleton. The alternate waves of extensions and contractions are aided by the leverage afforded by the buccal chamber and the setae.

or salivary gland cells and the glandular cells of the intestine are supposed to be the digestive glands which secrete digestive enzymes for digestion of food.

Respiratory System

The earthworm has no special respiratory organs like lungs or gills. Respiration takes place through the body wall. The outer surface of the skin is richly supplied with blood capillaries which aid in the diffusion of gases. Oxygen diffuses through the skin into the blood while carbon dioxide from the blood diffuses out. The skin is kept moist by mucous and coelomic fluid and facilitates exchange of gases.

Circulatory system

Lampito mauritii exhibits a closed type of blood vascular system consisting of blood vessels, capillaries and lateral hearts (Figure 4.4). Two median longitudinal vessels run above and below the alimentary canal as dorsal and ventral vessels of the earthworm. There are paired valves in the dorsal vessels which prevent the backward flow of the blood. The ventral vessel has no valves and is non contractile, allowing the backward flow of blood. In the anterior part of the body the dorsal vessel is connected with the ventral vessel by eight pairs of **commissural vessels** or the **lateral hearts** lying in the 6th to 13th segments. These vessels run on either side of the alimentary canal and pump blood from the dorsal vessel to the ventral vessel. The dorsal vessel receives blood from various organs in the body. The ventral vessel supplies blood to the various organs. Blood glands are present in the

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